

2. Method Declaration: The first line of a method that defines its name, return type and parameters.

Example: `public static int add(int a, int b)`

Method body: The code inside the `{ }` that runs when the method is called.

Example: `{
 return a + b;
}`

3. An access modifier keyword is used to change the access level of a method.

Examples:

- `public`
- `private`
- `Protected`

4. Visibility. This describes where the method can be accessed from.

5. `public class ScopeExample {`

```
    public static void method1()  
    {  
        int var3;  
        for (int var4 = 0; var4 < 5; var4++) {  
            method1();  
        }  
    }  
}
```

Explanation:

- `var1`: Only usable inside `main`
- `var2`: Only usable inside the `for` loop in `main`
- `var3`: Only usable inside `method1`
- `var4`: Only usable inside the `for` loop in `method1`

6. a) A class method named `getVowels` that can be called by any other method, requires a `String` parameter, and returns an integer value.

```
public static int getVowels(String text)
```

b) A class method named `extractDigit` that can be called by any other method, requires an integer parameter, and returns an integer value.

```
public static int extractDigit(int number)
```

c) A class method named insertString that can be called by any other method, requires a String parameter and an integer parameter, and returns a String parameter.

```
public static String insertString(String text, int position)
```

7. a) The compiler uses the method signature. This includes the method name, the number of parameters, types and orders of parameters.

b) Yes, this is method overloading. Two methods can have the same name as long as their parameter lists are different.

8. a) A return statement is used to send a value back to the method that called it and immediately stop the method's execution.

Example: return result;

b) Only one value can be returned at a time.

c) A method that returns a value uses a data type like int, double, String in its declaration. It must include a return statement that sends a value back.

Example: public static int add(int a, int b)

A method that does not return a value uses the keyword void in its declaration. It does not return anything so no value is sent back.

Example: public static void printMessage()

9. Error 1: The class is closed too early, so main is not inside the class. In Java, all methods must be inside a class.

Error 2: This is invalid syntax.

11. True or False:

- a) True. Breaking a problem into smaller methods is called procedural abstraction.
- b) False. A method call is when you use the method, not declare it.
- c) False. A void method does not return a value.
- d) False. An access modifier like public, private controls visibility, not return type.
- e) True. static means the method belongs to the class, not an object.
- f) False. Parameters are enclosed in parentheses (), not braces { }.
- g) False. Local variables can only be used inside the method they are declared in.
- h) True. You can change the parameter variable inside the method, but It does not affect the original argument for primitive types like int.
- i) False. Method overloading means having multiple methods with the same name but different parameters, not the number of methods.
- j) True. The return statement sends a value back to the calling method.
- k) False. A precondition describes what must be true before the method runs like valid input, not just data types.
- l) False. A postcondition describes the result/output after the method runs, not how it works.